Chanu Kalubowila

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Education

Bachelor of the Science of Engineering (Hons.) Aug. 2021 – Jul. 2025
University of Moratuwa, Sri Lanka (Washington Accord Accredited)

- Mechatronic Systems Engineering Stream; First Class
- Minor in Mathematics
- Ranked **9th** in Mechanical Engineering (120 students)

Teaching & Industrial Experience

Temporary Lecturer May 2026 – Present
Department of Mechanical Engineering, University of Moratuwa

- Lectures, labs, and workshops: PCB Designing, Instrumentation System Design, Industrial Automation, Control Systems, Mechatronic System Design Project, and MEMS.

Academic Instructor Nov. 2025 – May 2026
Department of Mechanical Engineering, University of Moratuwa

Trainee Mechatronics Engineer Dec. 2023 – Jun. 2024
Colombo International Container Terminals, Sri Lanka

Publications & Patents

Refereed Conference Papers

1. **Kalubowila K.D.C.S.**, Gayan K.A.K., Perera M.A.Y., Jayathilaka W.A.D.M., Amarasinghe Y.W.R. "Passive Mechanical Retention of Capillary Tubes for Machine Vision Compatible High Speed PCV Test Centrifugation Rotor." In: *Moratuwa Engineering Research Conference (MERCon)*, Sri Lanka, Aug. 2026. **Best Paper Award.** DOI: 10.1109/MERCon71835.2026.11691396
2. Perera M.A.Y., **Kalubowila K.D.C.S.**, Gayan K.A.K., Amarasinghe Y.W.R., Jayathilaka W.A.D.M. "Design and Manufacturing Optimization of a Highspeed PCV Centrifuge Machine." In: *MERCon*, Sri Lanka, Aug. 2025. DOI: 10.1109/MERCon67903.2025.11217024

Journal Papers (Under Review)

3. **Kalubowila K.D.C.S.**, et al. "Design and Clinical Validation of a Vision-Based Automated Centrifuge Instrument for PCV Measurement." *Measurement* (Elsevier).
4. Perera M.A.Y., **Kalubowila K.D.C.S.**, et al. "Automated Hematocrit Estimation from Multi-Tube Centrifuge Rotor Images via Oriented Detection and Axis-Constrained Boundary Regression." *Biomedical Signal Processing and Control* (Elsevier).
5. Gayan K.A.K., **Kalubowila K.D.C.S.**, Perera M.A.Y., Amarasinghe Y.W.R., Jayathilaka W.A.D.M. "Optimal PID Controller Design for Brushed DC Motors using CMA-ES Optimization." *Engineering Research Express* (IOP).

Patents & Registered Designs

- **Patent (Pending):** "A Rotor Disk in Centrifuge Machines for the Packed Cell Volume Test." App. No. **P/1/023851**, National Intellectual Property Office of Sri Lanka (NIPO).
- **Industrial Design (Registered):** "Centrifuge Machine." App. No. **14341**, University of Moratuwa.
- **Industrial Designs (Filed):** STM32H743 Flight Controller Carrier Board; Bare-Metal Sensorless FOC ESC Board.

Honours & Awards

Best Paper Award — MERCon 2026 (Mechanical Systems Track; Final Year Project) 2026
IESL Undergraduate Inventor of the Year 2025 — 1st Place (Institution of Engineers, Sri Lanka; Final Year Project) Jul. 2025
Mobile Award — Best Mechanical Engineering Final Year Project (McLarens Lubricants) 2025
Dean's List — Faculty of Engineering, University of Moratuwa (4 semesters (1, 2, 3 and 8)) 2021–2025
Baseball — University Colors & Inter-University Champions 2023, 2024

Research Projects

Vision-Based PCV Analyzer — Final Year Research Project Jan. 2024 – Jul. 2025
University of Moratuwa • *Portfolio*

Novel lidless passively-retaining centrifuge rotor and deep-learning segmentation pipeline for automated hematocrit measurement.

Flapping-Wing Robot — Full-Stack Design, Simulation & Prototype Aug. 2023 – Dec. 2024
Foundation of Cross-Domain Simulation Work • *Portfolio*

Fabricated a 110 cm, 4.3 N bio-inspired ornithopter from scratch and built custom ROS+Gazebo aerodynamic plugins as an open-source alternative to MATLAB/Simulink in flapping-wing research.

Engineering Projects

Designed and Fabricated a Custom Sensorless FOC ESC May 2026 – Present
STM32G431 • *Portfolio*

Built a 2-layer power stage (discrete gate drivers, optocoupler isolation, low-side shunt sensing, onboard buck regulation) driving low-inductance UAV motors via a 24 kHz sensorless FOC loop – State-Space Flux Observer and hardware CORDIC – in a 2,728-cycle (16.05 μs) control loop.

Designed and Flight-Validated a Bare-Metal Betaflight Flight Controller Jan. 2026 – Present
STM32H743 • *Portfolio*

Designed a 4-layer carrier board around a bare industrial H743 devboard and ported Betaflight onto it, root-causing three silicon-level clock/DMA bugs to reach stable flight on an 8-motor/4-servo quadcopter with dual-IMU redundancy.

PCB Digital Twin — Open-Source Plugin Sept. 2026 – Present
Independent Research • *Portfolio*

Built a real-time visualizer that renders a KiCad PCB as a living system (per-net currents, voltages, 2D thermal fields) driven by firmware on a virtual STM32 in Renode.

Lab-on-a-Chip Dielectrophoresis Device (MEMS) Jul. 2024 – Oct. 2024
Design of a Microfluidic Blood Particle Separation LOC

Autonomous Miniature Taxi Robot Mar. 2025 – Jul. 2025
PID Navigation + ML Sign Detection

Fundamental SCADA System Jul. 2024 – Oct. 2024
ESP32 + MySQL + Closed-Loop Thermal Control

IoT-Based Weather Monitoring System 2023
Embedded Sensor Network

Technical Skills

Programming: Python, C, C++, MATLAB, ROS1
Embedded: STM32 (F4/G4/H7), Renode, PlatformIO, STM32CubeIDE, ARM GCC
Simulation: ngspice, ANSYS, COMSOL, MSC Adams, Gazebo, MATLAB/Simulink
PCB Design: KiCad, Proteus, Gerber generation, BOM management
PLC/HMI: Siemens TIA Portal, Delta, HMI design
Control: PID, FOC, SMO, Active Flux Observer, CORDIC, State-Space Observers
Vision: YOLOv8-OBB, U-Net, PyTorch, OpenCV, FFT analysis
Hardware: Soldering, oscilloscope, logic analyzer, electrical wiring, microcontroller interfacing, arc/MIG welding

Extracurricular & Volunteer

Organizing Committee — MERS 2025 (Mechanical Engineering Research Symposium, UoM) Jan. 2026
Organizing Committee — MechatronX 2023 (EXMO 2023, UoM) Jul. 2023
Safety Committee — EXMO 2023 (UoM) Jul. 2023

References

Prof. Y. W. R. Amarasinghe
Program Coordinator
BSc Eng Hons in Mechatronics and Automation Systems Eng.
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University of Moratuwa, Sri Lanka
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Dr. W. A. D. M. Jayathilaka
Stream Coordinator
Mechatronics and Automation Systems Eng.
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